Affordable Solar Radio Telescope

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Abstract: The Affordable Solar Radio Telescope is a radio telescope operating in the Ku band which extends from 12 to 16 GHz. As the name suggests, the cost of the radio telescope is minimal and it is designed such that it can be replicated by anyone and everyone. The cost of building has been kept low by using off the shelf equipment which is easily available anywhere in the world. The telescope is primarily intended to observe the sun in this band viz. 10.7 to 12.75 GHz. but can just as easily be used for observing radiation from compact fluorescent lamps, human body, boiling water etc. as well. Although the radio telescope is easy to build and operate, that doesn't limit the science that can be done with it as it can be used to carry out several scientific endeavors both basic as well as advanced.

1 Aim

1. We give a step by step instructions for assembling the ASRT.
2. To get familiarize with the different co-ordinate systems used in Radio Astronomy.
3. Radio Intensity of the Sun in the given frequency range is to observed.
4. To get signal from various Geostationary satellites working in the Radio frequency range.

2 Apparatus Required

Ku Band Dish Antenna, Set Top Box(Optional), Satellite Finder, LNB, Co-axial Cable, F-type Connectors and Crocodile Clips, Mounting Stand and LNB Arm, Digital Multimeter, Pliers, Screw driver, Compression type Crimping tool, Spanner, Cable Preparation tool

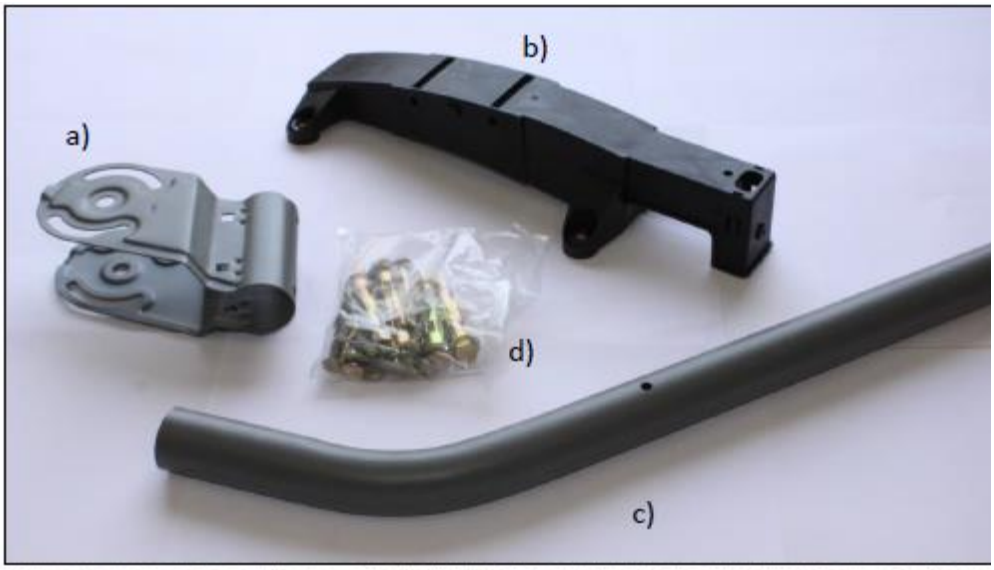
3 Set-up procedure

3.1 Components

The commercially available DTH dish comes with following components for mounting. The various components are: a) Mounting Bracket b) Dish Holder c) Dish Arm (which will not be used!) d) Nut-bolts, Screws, Washers, etc. e) LNB Arm f) LNB Holder as shown in the diagram.

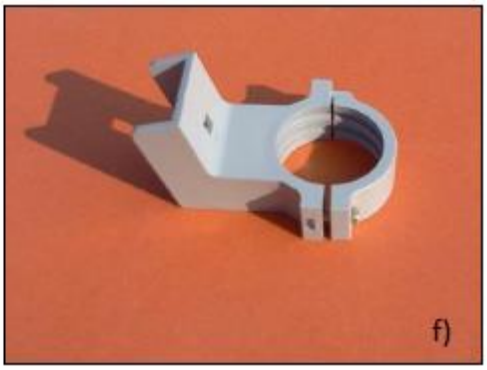
3.2 Dish Assembly

Step 1: Insert the Mounting Bracket over the Dish Holder such that it clicks into the central stub of the Dish Holder. Ensure that the Mounting Bracket is the right side up (if it is oriented properly it should slant down when held with the mounting clamp roughly perpendicular to ground as shown – the part held in hand which clamps on to the stand is the mounting clamp.) Put the appropriate Nut bolts as shown but do not tighten them too much (don’t forget to put the washers in before the nuts go on the bolts).


a)
b)
c)
d)

UPSIDE
e)

f)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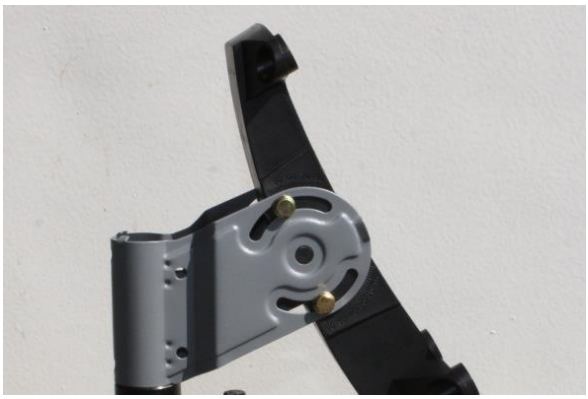
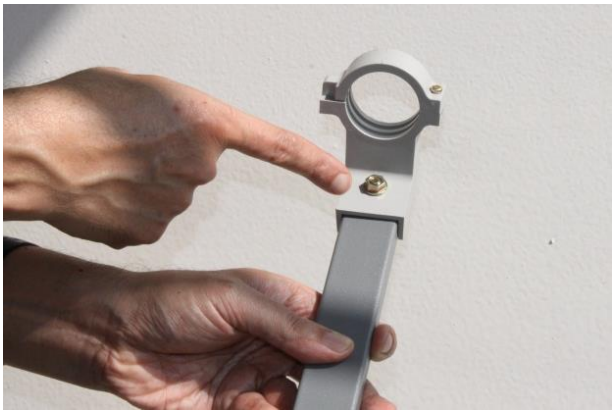


Step 2: Attach the dish holder to the dish using screws as shown. All the four nuts of the dish holder should be aligned properly with the fitting holes on the dish. After aligning the holder, simply screw in the appropriate screws (shown below) until they are sufficiently tight to hold the dish firmly.

Step 3: The dish can now be mounted on the stand by inserting the Mounting Bracket's clamp into the stand axle. Insert the LNB Arm, with the letters "UPSIDE" facing up, carefully into its holder at the bottom of the dish. Push it all the way in and ensure that it has clicked into place in the Dish Holder.

Note: Once the LNB Arm is in place do not try to remove it. The arm locks into place with the help of plastic locks. These can easily get damaged if the arm is pulled out after locking.
Also care should be taken not to put any unnatural stresses on the LNB arm to prevent it from bending and losing its alignment with the dish's focus.
Step 4: Connect the LNB Holder to the LNB Arm. Secure it with the appropriate screw. Ensure that the washer and nut is facing up and not the Bolt head.



Step 5: Loosen the LNB Holder screw and insert the LNB with the Feed side facing towards the dish into it. Tighten the screw so that the LNB stays in place but allow for some slack so that the LNB can be rotated within the holder.

This completes the dish assembly.

3.3 Cable connections

Before the connections are done the connectors need to be crimped onto the cables. There are two cables – one 5 meter long and the other 3 meters. The longer cable (5 m.) is used to connect the LNB to the Satellite Finder. The shorter cable (3 m.) connects the Satellite Finder to the battery.
The **crimping procedure** is available at
https://www.youtube.com/watch?v=fVtaRWiK5jo

Step 1: Take one piece of cable (5 m.) and connect it to the LNB of dish. Connect the other end of the cable to the Satellite Finder. Ensure that this end is connected to the "LNB IN" connector only. This is marked on underside of the Satellite Finder.

Step 2: Take the smaller cable (3 m.) and connect one end to the battery with the help of crocodile clips. Connect the other end to the Satellite Finder at the "TO REC" connector marked on the underside of Satellite Finder.

Fig. The pictures show the dish with the LNB Arm connected upside down. The LNB Arm should be connected such that the “UPSIDE” letters face up towards the sky after connecting it to the dish.

4 Theory

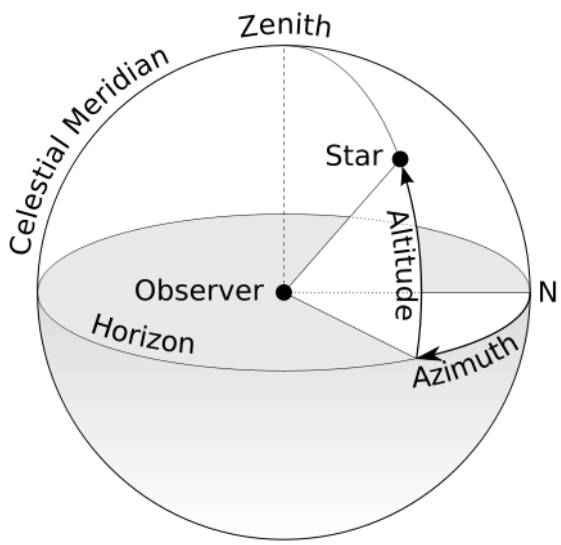
The visible and radio skies reveal quite different "parallel universes" sharing the same space. Most bright stars are undetectable at radio wavelengths, and many strong radio sources are optically faint or invisible. Familiar objects like the Sun and planets can look quite different through the radio and optical windows. The extended radio sources spread along a band from the lower left to the upper right in this picture lie in the outer Milky Way. Unlike the nearby (distances 1000 light years) stars visible to the human eye, almost none of the myriad "radio stars" (unresolved radio sources) scattered across the sky are actually stars. Most are extremely luminous radio galaxies or quasars, and their average distance is over 5,000,000,000 light years. Radio waves travel at the speed of light, so distant extragalactic sources appear today as they actually were billions of years ago. Radio galaxies and quasars are beacons carrying information about galaxies and their environs, everywhere in the observable universe and ever since the first galaxies were formed.

The brightest discrete radio source is the Sun, but it is much less dominant than it is in visible light. The radio sky is always dark, even when the Sun is up, because atmospheric dust doesn't scatter radio waves, whose wavelengths are much longer than the dust particles.

To set our radio telescope to view the Radio Sun and Geo-stationary satellites, we need to know about co-ordinate systems used in Radio Astronomy. The current position of Sun and Moon is obtained using http://www.satellite-calculations.com/Satellite/suncalc.htm

4.1 Horizontal Coordinate System

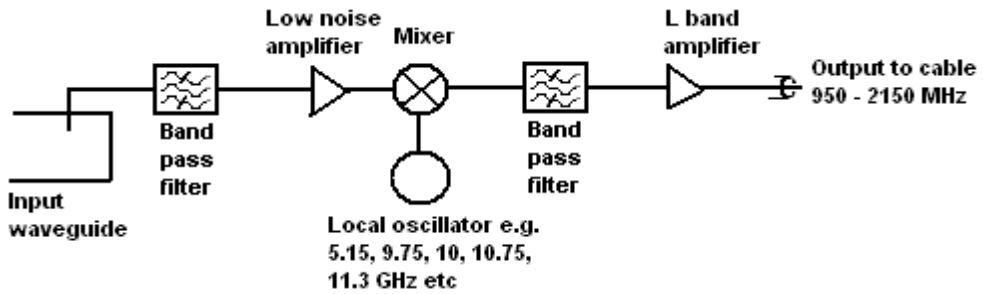
The horizontal coordinate system is a celestial coordinate system that uses the observer's local horizon as the fundamental plane. This coordinate system divides the sky into the upper hemisphere where objects are visible, and the lower hemisphere where objects cannot be seen since the earth is in the way. The great circle separating hemispheres is called **celestial horizon or rational horizon**. The pole of the upper hemisphere is called the zenith. The pole of the lower hemisphere is called the **nadir**.



The horizontal coordinates are:
Altitude (Alt), sometimes referred to as elevation, is the angle between the object and the observer's local horizon. For visible objects it is an angle between 0 degrees to 90 degrees.
Azimuth (Az), that is the angle of the object around the horizon, usually measured from the north increasing towards the east.
The horizontal coordinate system is sometimes also called the Alt/Az system, or the altazimuth system (from the name of the altazimuth mount for telescopes, whose two axes follow altitude and azimuth).
A video explanation of this system is available at
http://www.youtube.com/watch?v=nA8CCqA-JlY

4.2 LNB Characterization

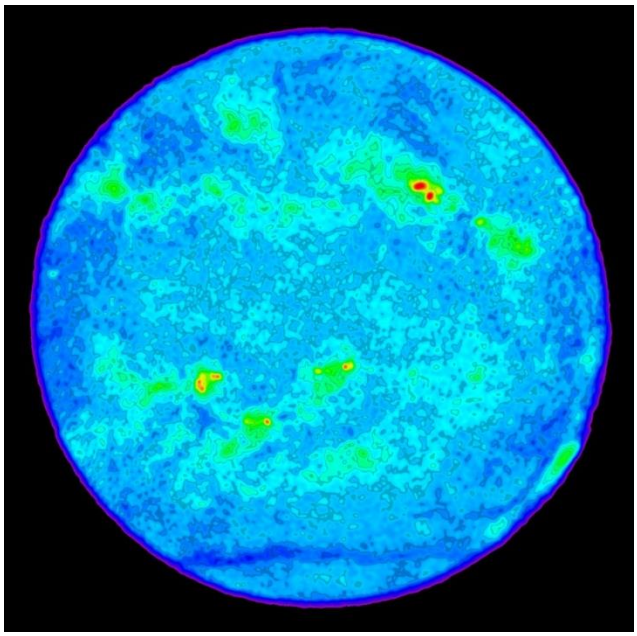
LNB is an acronym for Low Noise Block converter. Each receiver head normally consists of one or more LNB receiver units with its own local oscillator, mixer, amplifier and filters. A functional diagram is presented in Figure 1. .Each LNB accepts signals over a broad band from about 11.5 to 13.2 GHz and this block is mixed with a local oscillator signal of about 11 GHz. Thus the output block contains a range of Intermediate Frequency (IF) signals extending from about 0.3 to 2 GHz.



The LNB is the device on the front of a satellite dish that receives the very low level microwave signal from the satellite, amplifies it, changes the signals to a lower frequency band and sends them down the cable to the indoor receiver.
The LNB used in our setup is a Ku-Band LNB with Input Band 10.95 - 12.15 GHz, Local Oscillator Frequency 10 GHz and Output L Band frequency 950-2150 MHz.

4.3 Facts About Radio Sun

The radio Sun is somewhat bigger than the optical Sun: the solar limb (the edge of the disk) is about 20000 km above the optical limb.
The quiet Sun at frequency 46 GHz imaged by the VLA with a resolution of 12 arcsec, or about 8400 km on the surface of the Sun. The brightest features (red) in this false-color image have brightness temperatures T_b =106 K and coincide with sunspots. The green features are cooler and show where the Sun's atmosphere is very dense. At this frequency the radio-emitting surface of the Sun has an average temperature of $3x10^4$ K, and the dark blue features are cooler yet. The blue slash crossing the bottom of the disk is a feature called a filament channel, where the Sun's atmosphere is very thin: it marks the boundary of the South Pole of the Sun.



5 Data and Evaluation

5.1 Measurement of Radio Sun

Measurements are taken at 3 pm from NISER terrace on 3-12 December, 2014.

Azimuth	Altitude	Left Point for start of signal	Right Point of drop of signal	Angle (in deg)
240˚	21˚	65	110	45
240˚	16 ˚	66	111	45
240˚	14 ˚	64	110	46
240˚	8 ˚	66	109	43
239˚	25 ˚	65	110	45
239˚	23 ˚	67	111	44
239˚	18 ˚	67	110	43

5.2 List of Satellites Detected

Name of Satellite	Alt/Azi	Intensity(in local units)
INSAT 4B	Azimuth=138.9° SE Altitude= 59.9°	235
INSAT 3A	Azimuth=139.1° SE Altitude= 59.9°	220
INSAT 3E	Azimuth=109.3° ESE Altitude= 34.0°	230

Information about the detected satellites:

1) INSAT 3A Launched: 09 Apr 2003

Brightness: 3.0 mag (at 1000 km, 50% illuminated)
10.0 mag (at perigee, full illumination)
Mean magnitude from radar observations

Orbit: 35776 x 35799 km, 23.94h Inclination: 0.1°

Geostationary at 93.5° geographical Longitude

2) INSAT 3E Launched: 27 Sep 2003

Brightness: 3.0 mag (at 1000 km, 50% illuminated)
10.0 mag (at perigee, full illumination)
Mean magnitude from radar observations

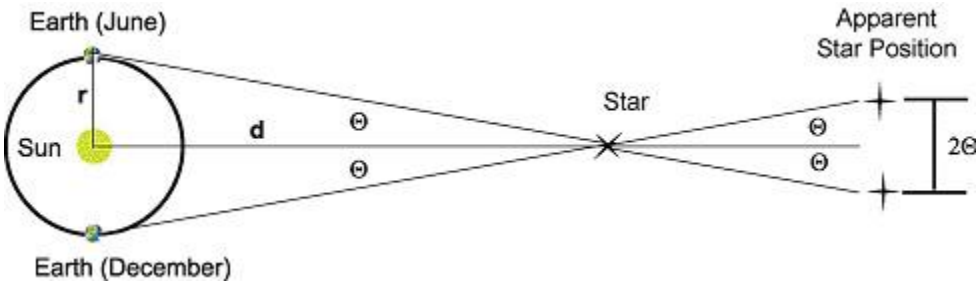
Orbit: 35561 x 35719 km, 23.81h Inclination: 0.6°

3) INSAT 4B
   Launched: 11 Mar 2007
   Brightness: 3.0 mag (at 1000 km, 50% illuminated)
   10.0 mag (at perigee, full illumination)
   Mean magnitude from radar observations
   Orbit: 35763 x 35812 km, 23.94h Inclination: 0.1°
   Geostationary at 93.5° geographical Longitude

6 Calculations

We have, 1 A.U = 149 597 871 km
Optical Radius of Sun = 695,800 km

Using method of parallax:



We have,

Angle	tan(θ)	Radius (km)
45	0.557852	4172671.63
45	0.557852	4172671.63
44	0.433266	3240786.1
43	0.534864	-4000728.8
45	0.557852	4172671.63
44	0.433266	3240786.1
43	0.534864	-4000728.8
46	0.433266	3240786.1

7 Error Analysis

Theoretical value of Solar radius, R = 695,800 km Size = 1391600 km
Relative Error,
$\Delta R = R - R_{obs}/R$

Radius (km)	ΔR (km)
4172671.63	-4.99694
4172671.63	-4.99694
3240786.1	-3.65764
-4000728.8	6.749826
4172671.63	-4.99694
3240786.1	-3.65764
-4000728.8	6.749826
3240786.1	-3.65764

8 Results

The Solar Sun is observed to be bigger in size as expected. The estimated values are well within experimental error of ±5%.

9 Conclusions and Discussions

- The ASRT was setup from scratch and calibrations were done carefully.
- One of the unusual things noticed were a hierarchy of the radio intensities for the following:
  SUN > SATELLITES > MOON > HUMAN
  The readings were a bit crude so we don't mention them here but the observation was strict.
- Also calibration of the amplifier that we use was a crude idea which can be made better in other ways.
- Radio spectrum of different sources turns out to be interesting and can be pursued further next semester.
- The results for the smaller scattering angles do not correspond well with the theoretical prediction they nevertheless show that with decreasing scattering angle, the difference in wavelength also decreases.
- The error in measurement is also due to the dead time of the GM counter for which necessary counting corrections are introduced.